

Adam Pardyl

PhD Candidate in CS, ML/AI Researcher, Software Engineer.

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EDUCATION

- **Jagiellonian University** Kraków, Poland
PhD candidate in Computer Science Oct. 2022 – Oct/Nov. 2026
 - Published at **NeurIPS, ECCV, IJCAI, WACV**; current GPA 5.0/5.
 - Thesis topic: "*Efficient computer vision for embodied agents*", focus on vision-action and vision-language-action models.
 - Researching transformer based predictive and generative methods for embodied agents, with self-supervised pre-training and fine-tuning using reinforcement learning for quick scene understanding and world representation learning.
- **Jagiellonian University** Kraków, Poland
Graduated with honors as Master of Computer Science Oct. 2020 – Jun. 2022
 - Specialization: **Machine Learning**; GPA 4.9/5; Co-authored a research paper published at ECML PKDD.
- **Jagiellonian University** Kraków, Poland
Graduated as Bachelor of Computer Science – IT Analyst Oct. 2016 – Jun. 2019
 - Specialization: **Theoretical Computer Science**; GPA 4.0/5; Participated in an interval scheduling research project.
 - Focus on algorithms, formal methods, discrete mathematics and probability theory.
- **The August Witkowski 5th High School** Kraków, Poland
Sep. 2013 – May 2016
 - Finalist of **Polish Olympiad in Informatics (2016)**.
 - First place in *Polish Electrical and Electronic Knowledge Olympiad* in ICT category (2016).
 - Second place in *2015 European CanSat Competition* organized by the European Space Agency.

SELECTED PUBLICATIONS

- FlySearch: Exploring how vision-language models explore. Pardyl, A.; Matuszek, D.; Cygan, M.; et. al. Conference on Neural Information Processing Systems (**NeurIPS**) 2025.
 - Introduced a **spatial reasoning benchmark for VLMs** with realistic 3D outdoor environments using Unreal Engine 5.
- AdaGlimpse: Active Visual Exploration with Arbitrary Glimpse Position and Scale. Pardyl, A.; Wronka, M.; Wołczyk, M.; et. al. In European Conference on Computer Vision (**ECCV**) 2024.
 - Created a **reinforcement learning method for fast, human-like visual environment exploration** and representation.
- Beyond Grids: Exploring Elastic Input Sampling for Vision Transformers. Pardyl, A.; Kurzejamski, G.; Olszewski, J.; et. al. Winter Conference on Applications of Computer Vision (**WACV**) 2025.
 - Devised and performed a study on **token-efficient, grid-free image sampling** techniques for vision transformers.
- Active Visual Exploration Based on Attention-Map Entropy. Pardyl, A.; Rypeść, G.; Kurzejamski, G.; et. al. In International Joint Conference on Artificial Intelligence (**IJCAI**) 2023.
 - Introduced a **most informative next observation prediction** method, leveraging **uncertainty estimation** for MAE.

WORK EXPERIENCE

- **IDEAS NCBR (soon to be IDEAS Research Institute)** Warsaw, Poland
PhD student researcher (as part of the PhD program) Sep. 2022 – present
 - **Lead investigator for visual exploration project.** Researching vision transformer-based models for embodied AI, dedicated to efficient visual environment exploration and scene understanding (as part of PhD thesis).
 - Mentored 2 MSc students resulting in successful PhD program admissions.
- **AGH University of Krakow** Krakow, Poland
Machine Learning Engineer Sep. 2021 – Dec. 2022
 - Implemented a **deep learning-based diagnostic pipeline for early detection of lung cancer** using CT scans, outperforming human performance on both publicly available benchmarks and an in-house dataset.
 - Work done as part of the National Centre for Research and Development project "X-rAI: Diagnostic browser for radiology with computer aided engineering using Artificial Intelligence". Results released as a product.
- **Google** Zürich (remote), Switzerland
Software Engineering Intern Jul. 2020 – Oct. 2020
 - Worked with the YouTube Content ID team on improving content matching system.
 - Performed big data analysis to identify highly repetitive, yet non-distinctive video fragments on YouTube in order to **optimize video matching subsystem improving its performance**.
 - Designed an ML-based detector to reduce the number of false-positive Content ID early matches by over **82%**.

- **Google** Zürich, Switzerland
Software Engineering Intern Jul. 2019 – Sep. 2019
 - Joined the Android Frameworks team working on the Android Operating System.
 - Implemented a new high-performance on-device binary logging solution for the Android Window Manager, utilizing compile-time Java code generation, enabling verbose analytics and zero overhead in production.
- **Google** London, UK
Software Engineering STEP Intern Jul. 2017 – Sep. 2017
 - Worked with the Android for Corp team (part of Android Enterprise).
 - Developed and launched a new security feature for the G Suite Mobile Management platform, enabling enterprise administrators to monitor and mitigate potentially harmful apps on corporate devices.

PATENTS

- (pending) A computer implemented method for identifying a microorganism in a blood and a data processing system for therefore. European Patent Office, 2024, Application number: EP24461637.1.

RESEARCH GRANTS

- **(primary investigator) National Science Center Preludium Grant:** Where to look next - guiding active visual exploration with internal model uncertainty.
- **(participant) National Science Center Sonata bis Grant:** Sustainable computer vision for autonomous machines.
- **(participant) Foundation for Polish Science Team-Net Grant:** Bio-inspired artificial neural networks.

PHD SUMMER SCHOOLS

- International Computer Vision Summer School (ICVSS) 2025
- ELLIS Doctoral Symposium 2024 & 2025
- MLSS^D Machine Learning Summer School on Drug and Materials Discovery 2025
- Mediterranean Machine Learning (M2L) Summer School 2023 & 2025
- MLSS^S Machine Learning Summer School on Applications in Science 2023

LEADERSHIP & ACTIVITIES

- **Computer Science Students' Association at the Jagiellonian University** Kraków, Poland
Vice-president of the Board, President of the Revisory Committee; Software Developer Oct. 2017 – Oct. 2022
 - Managed the work of the most successful student organization at the Faculty of Mathematics and Computer Science of the Jagiellonian University, increasing the participation by 60% up to 140 members.
- **TechSwarm 2015** Kraków, Poland
Team Leader; Software Developer Aug. 2014 – Jul. 2015
 - Built a miniature space probe model, capable of measuring various parameters of planet's surface and transmitting them to a receiving station. Second prize in the 2015 European CanSat Competition organized by the *European Space Agency*.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, JavaScript, SQL, basics of CUDA and x86_64 Assembly.
- **Deep Learning & Vision:** PyTorch, PyTorch Lightning, Hugging Face Transformers, OpenCV, Unreal Engine 5.
- **Infrastructure & MLOps:** Linux (Debian/Arch), Docker, Slurm, WandB, Git, TCP/IP networking, RPC.

RESEARCH INTERESTS

- **Vision-language-action models:** leveraging natural language for action planning and spatial reasoning.
- **Human-like world perception:** reasoning based on environmental cues and multi-modal contextual information.
- **Computer vision for embodied AI:** vision-guided autonomous agents in open world problems, intuitive exploration.

LANGUAGES

- **Polish:** Native
- **English:** Full Professional Proficiency